

3.13

Let X be the number of correctly answered questions. Then $X \sim \text{Bin}(20, 0.2)$. From the binomial distribution table,

$$P(4 \leq X \leq 8) = F(8) - F(3) = 0.9900 - 0.4114 = 0.5786$$

It can also be obtained using R by

`>pbinom(8, 20, 0.2)-pbinom(3, 20, 0.2)`

3.22

$X \sim \text{Bin}(3, 1/3)$

(a)

$$f(0) = \left(\frac{2}{3}\right)^3 = 0.2963, \quad f(1) = 3 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^2 = 0.4444,$$

$$f(2) = 3 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = 0.2222, \quad f(3) = \left(\frac{1}{3}\right)^3 = 0.0370$$

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 0.2963 & \text{if } 0 \leq x < 1 \\ 0.2963 + 0.4444 = 0.7407 & \text{if } 1 \leq x < 2 \\ 0.2963 + 0.4444 + 0.2222 = 0.9630 & \text{if } 2 \leq x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$

(b) $f(1) + f(3) = 0.4444 + 0.0370 = 0.4814$

(c) $np = 3 \left(\frac{1}{3}\right) = 1$

(d) $np(1 - p) = 3 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right) = \frac{2}{3}$

3.28

(a)

$$\begin{aligned} E(X) &= \sum_{x=0}^5 xf(x) \\ &= 0(0.2) + 1(0.32) + 2(0.25) + 3(0.16) + 4(0.04) + 5(0.03) \\ &= 1.61 \end{aligned}$$

$$\begin{aligned} E(X^2) &= \sum_{x=0}^5 x^2 f(x) \\ &= 0(0.2) + 1(0.32) + 4(0.25) + 9(0.16) + 16(0.04) \\ &\quad + 25(0.03) \\ &= 4.15 \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 4.15 - [1.61]^2 = 1.5579$$

(b) Let Y be the event that a text contains at most one typo. Then $Y \sim \text{Bin}(10, p)$, where

$$p = P(X \leq 1) = f(0) + f(1) = 0.52.$$

The probability that 6 of these texts contain at most one type is

$$P(Y = 6) = \binom{10}{6} (0.52)^6 (0.48)^4 = 0.2204.$$

(c) $E(X) = np = 10(0.52) = 5.2$

$$\text{Var}(X) = np(1 - p) = 10(0.52)(0.48) = 2.496$$

3.35

Let X be the number of hard cover texts. Then X has a hypergeometric distribution with

$$P(X = x) = \frac{\binom{25}{x} \binom{75}{10-x}}{\binom{100}{10}}, x = 0, 1, \dots, 10.$$

(a)

$$P(X = 2) = \frac{\binom{25}{2} \binom{75}{8}}{\binom{100}{10}} = 0.2924$$

(b) X is approximately $\text{Bin}(n, p)$, where

$$p = \frac{a}{N} = \frac{25}{100} = \frac{1}{4}, \quad n = 10.$$

$$P(X = 2) \approx \binom{10}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^8 = 0.2816$$

3.40

$$P(X = x) = \frac{5^x e^{-5}}{x!}, x = 0, 1, 2, \dots$$

(a)

$$P(X = 2) = \frac{5^2 e^{-5}}{2} = \frac{25}{2} e^{-5} = 0.0842$$

(b)

$$P(X \leq 2) = \sum_{x=0}^2 \frac{5^x e^{-5}}{x!} = \left(1 + 5 + \frac{25}{2}\right) e^{-5} = \frac{37}{2} e^{-5} = 0.1247$$

(c) $E(X) = \lambda = 5$ (d) $Var(X) = E(X) = 5$ **3.43**

Let X be the number of cars with a defective muffler. Then $X \sim \text{Bin}(400, 0.01)$. In Poisson approximation, $\lambda = np = 400(0.01) = 4$. Thus,

$$P(X = x) = \frac{4^x e^{-4}}{x!}, x = 0, 1, 2, \dots$$

(a) $P(X > 5) \approx 1 - F(5) = 1 - 0.7851 = 0.2149$ (b) $P(3 \leq X \leq 6) \approx F(6) - F(2) = 0.8893 - 0.2381 = 0.6512$ **3.47**

Let X denote the number of attempts needed to produce a craft item without failure. Then X is geometric with $p = 0.5$ and the probability distribution is given by

$$P(X = x) = (0.5)^{x-1}(0.5) = (0.5)^x, x = 1, 2, \dots$$

The probability is

$$P(X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.5 + (0.5)^2 + (0.5)^3 = 0.875.$$

3.53(a) $> \text{dgeom}(2, 0.4)$

0.144

(b) $> \text{pgeom}(2, 0.4)$

0.784

3.58

$$80 = \mu - 100, \quad 280 = \mu + 100, \quad k\sigma = 100 \Rightarrow k = \frac{100}{25} = 4$$

$$P(|X - \mu| < 100) = P(|X - \mu| < 4\sigma) \geq 1 - \frac{1}{4^2} = \frac{15}{16} = 0.9375$$

The probability is at least 0.9375.

3.61

Let X_1 denote the number of white persons you met, X_2 denote the number of blacks you met and X_3 denote the number of Asians you met in 2000. Then (X_1, X_2, X_3) has a multinomial distribution with probability distribution

$$f(x_1, x_2, x_3) = \frac{5!}{x_1! x_2! x_3!} (0.8)^{x_1} (0.08)^{x_2} (0.12)^{x_3}, \quad x_1, x_2, x_3 = 1, 2, \dots, 5, \\ x_1 + x_2 + x_3 = 5$$

(a)

$$f(3, 1, 1) = \frac{5!}{3!} (0.8)^3 (0.08)(0.12) = 0.0983$$

(b)

$$\begin{aligned} & f(3, 1, 1) + f(3, 2, 0) + f(3, 0, 2) + f(4, 1, 0) + f(4, 0, 1) + f(5, 0, 0) \\ &= \frac{5!}{3!} (0.8)^3 (0.08)(0.12) + \frac{5!}{3! 2!} (0.8)^3 (0.08)^2 \\ &+ \frac{5!}{3! 2!} (0.8)^3 (0.12)^2 + \frac{5!}{4!} (0.8)^4 (0.08) + \frac{5!}{4!} (0.8)^4 (0.12) \\ &+ (0.8)^5 \\ &= 0.0983 + 0.0328 + 0.0737 + 0.1638 + 0.2458 + 0.3277 \\ &= 0.9421 \end{aligned}$$

Alternatively, we can consider the marginal distribution as binomial. If X is the random variable denoting the number of whites out of the 5 people, then $X \sim \text{Bin}(5, 0.8)$ and the answer to this question is

$$\begin{aligned} P(X \geq 3) &= \binom{5}{3} (0.8)^3 (0.2)^2 + \binom{5}{4} (0.8)^4 (0.2) + (0.8)^5 \\ &= 0.2048 + 0.4096 + 0.3277 = 0.9421. \end{aligned}$$